

Convergence of recursive functions

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1 Introduction

Nowadays, stability is one of those things everyone is willing to achieve. Predictability, order, no chaos. And the behavior of any object can be expressed using functions. The simple functions we use on our daily basis are hardly ever stable: they output a different graph if we recursively apply a function inside itself. They also tend to behave chaotically if you repeat this process many times. Or do they? In this paper we will search for stability from the functions perspective and deduce how to find the stable intervals after applying the recursion.

The main goal of this paper is to investigate in a general way the convergence of recursive functions and find their basins of attraction. We will develop a method that accomplishes this goal and follow up with 3 examples to find useful applications.

The exponential family of functions will relate recursion to power towers. This will be helpful to study the convergence of power towers using our method. We will find out what is the result of raising a complex number to the power of itself infinitely many times: $z^{z^{z^{\dots}}}$. Does it converge to an imaginary number and therefore achieve stability? Or does it behave chaotically forever?

Definitions

- Definition 1: The functions analyzed in this paper will be represented as $f(x)$, their domain as D and their range as R . $f(x)$ will be continuous and differentiable on D . This condition rules out many unnecessary exceptions that exceed the aim of this paper.
- Definition 2: Representation of a recursive composition of the same function - $f^0(x) = x$ and $f^n(x) = f(f^{n-1}(x))$.
- Definition 3: Fixed point - For any function $f(x)$, a point satisfying $f(x_0) = x_0$.
- Definition 4: Attracting point [8] - A fixed point x_0 is called attracting if there exists

$\delta > 0$ such that for all x satisfying $|x - x_0| < \delta$,

$$\lim_{n \rightarrow \infty} f^n(x) = f(x_0) = x_0.$$

- Definition 5: Basin of attraction - The interval of values around x_0 , such that for every value,

$$\lim_{n \rightarrow \infty} f^n(x) = x_0.$$

- Definition 6: Tetration - ${}_n k = \underbrace{k^{k^{\cdot^{\cdot^{\cdot^k}}}}_n$. The repeated exponentiation from top to bottom. It is very important not to confuse the order of exponentiation in tetration:

$$\begin{cases} {}^3 k = k^{(k^k)} \\ {}^3 k \neq (k^k)^k \end{cases}$$

For example, for $k = 3$,

$$\begin{cases} {}^3 3 = 3^{(3^3)} = 3^{27} \\ {}^3 3 \neq (3^3)^3 = 3^9 \end{cases}$$

Claim 1 Let x_0 be a fixed differentiable point on the domain D . In order for x_0 not only to be a fixed, but also an attracting point, x_0 must satisfy $|f'(x_0)| < 1$.

This claim is proven in the appendix 5. So the final condition for an attracting point is,

$$\boxed{|f'(x_0)| < 1} \tag{1}$$

Initially, we could ask ourselves why does a derivative appear when talking about convergence. To understand what is happening as we iterate the function close to x_0 , let's analyze a graph. If we zoom in at any function infinitely close to a fixed point x_0 , the function becomes a straight line with gradient a . let's recursively apply $f(x)$ to $x_0 - \epsilon$: $f^n(x_0 - \epsilon)$. For my

visual example I use $n = 43$ as it is sufficiently big. For this graphical method I got inspired by the intuitive explanation by Dr. Grant Sanderson [10]:

1. Start at the point $P_0 = (x_0 - \epsilon, x_0 - \epsilon)$.
2. For $k \geq 0$, consider the vertical line V_k through P_k . Let Q_k be the intersection between V_k and f .
3. Consider the horizontal line H_k through Q_k . Let P_{k+1} be the intersection between H_k and $y = x$. This returns from the image of the function to its domain.
4. Repeat steps 2 and 3 for $k = 0, 1, \dots, n - 1$.

Applying this method for the gradient $|a| < 1$, we can see how on the figure 1a the function converges to x_0 . Adversely, when the gradient $|a| > 1$, the figure 1b shows how the function does not converge.

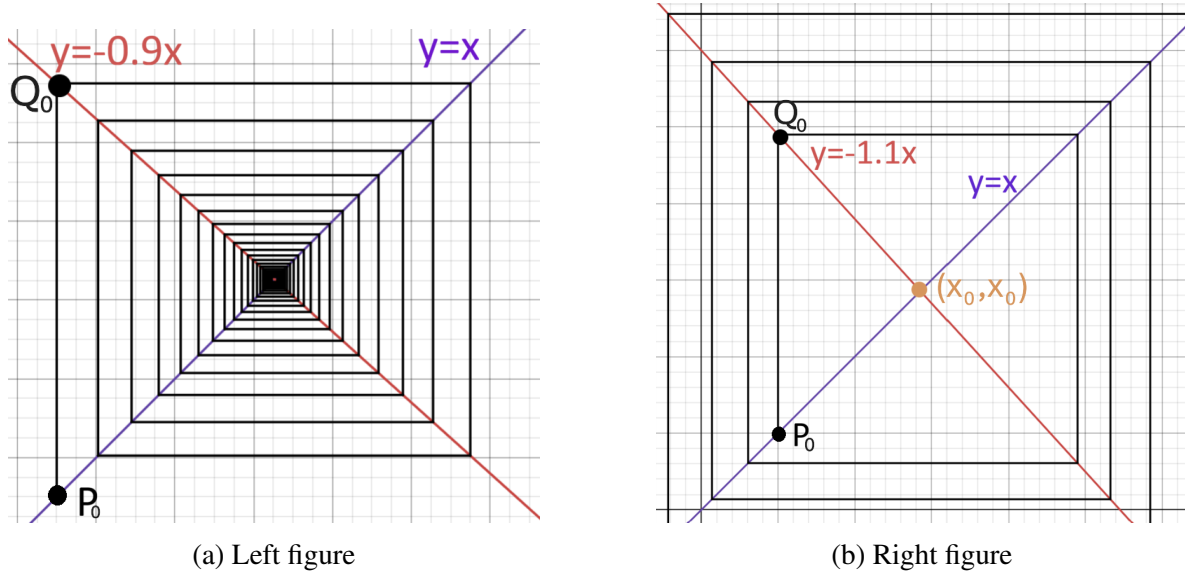


Figure 1: Graphical method

We have seen algebraically and geometrically how differs the behavior of the function near a fixed point x_0 , based on the slope of the function at x_0 .

1.1 Basin of attraction

Now, let's focus on the attracting point case, when $|f'(x_0)| < 1$. We will investigate what interval of values around x_0 that converges to x_0 .

Proposition

Let $f(x)$ be a function on the interval and domain D . If $|f'(x)| < 1$, and there is a fixed point x_0 , then $f(x)$ converges to a single point x_0 . In other words, $\lim_{n \rightarrow \infty} f^n(x) = x_0$.

Proof [6]

Let $x_1, x_2 \in D$ and $x_1 < x_2$. Since D is an interval and $f(x)$ is differentiable, by the Mean Value Theorem [1] there exists $\xi \in (x_1, x_2)$ such that

$$f(x_2) - f(x_1) = f'(\xi)(x_2 - x_1).$$

Taking absolute values,

$$|f(x_2) - f(x_1)| = |f'(\xi)| |x_2 - x_1|.$$

There exists $k \in (0, 1)$ such that any x_1 and x_2 we choose

$$|f'(\xi)| |x_2 - x_1| \leq k |x_2 - x_1|.$$

Thus, $f(x)$ is contracting. By the Banach fixed point theorem [7], there exists one single fixed point $x_0 \in D$, such that for every $x \in D$, $f^n(x)$ converges to x_0 as $n \rightarrow \infty$. So the 2 conditions for $f(x)$ to converge to x_0 are

$$\exists x_0 \in D \text{ such that } f(x_0) = x_0 \tag{2}$$

$$|f'(x)| < 1, x \in D \tag{3}$$

1.2 Basin Detection Method (BDM)

Now, we will come up with a method that we will use to find the attracting points and their ranges of convergence for any given function $f(x)$.

1. Optional refinement: applicable when $R \subsetneq D$. After the first iteration, we get R as the new domain for the second iteration. This means that the entire domain gets concentrated inside R . For that reason, if the convergence is true for f on the domain R , it will also be true for all domain D . For this reason, we can shrink D and consider only R . The motivation for doing this lies in increasing the basin of attraction near fixed points.
2. Solve $|f'(x)| < 1$ and recall D as the domain satisfying this equation. Now $f(x)$ satisfies the condition 3. D is made up of a set of n continuous intervals: $D = \{I_1, I_2, \dots, I_n\}$, $n \in \mathbb{N}$. The next step could be combined with this step requiring to solve a system of equation, but they can also be computed separately.
3. If possible, solve the equation $f(x) = x$ to find all the fixed points. The solution for this equation will be m points with $m \in \mathbb{N}$, $m \leq n$. If the equation is hard to solve, draw the line $x = y$ and find its m intercepts with $f(x)$. Note that we have already applied the slope condition 3, so each of the fixed points will also be an attracting point. In the case that $m = 0$, the function does not converge to any value.
4. Eliminate from D all the intervals that do not contain a fixed point. We are left with $D = \{I_1, I_2, \dots, I_m\}$.

For each interval I_k of D , $1 \leq k \leq m$, the domain is D_k and the range - R_k . Note that $|f'(x)| < 1$ for all $x \in I_k$, and there is a fixed point $x_k \in I_k$. These are exactly the 2 conditions we needed so $f^n(x)$ converges to x_k as proven at 1.1. Therefore, $f(x)$ has m different attracting points and each interval I_k converges to x_k . This process could be applied

to any continuous and differentiable function to find the stable points and their basins of attraction. Figure 2 shows an example of the visual application of our BDM on a polynomial function.

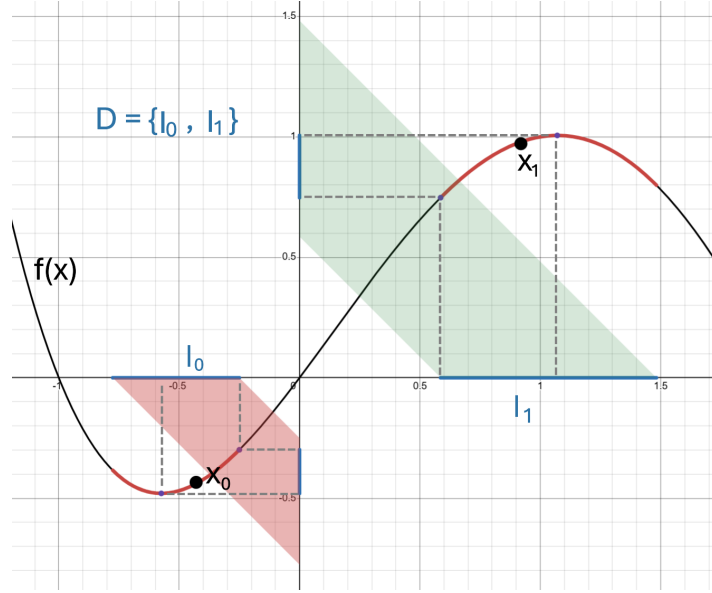


Figure 2: Set of intervals I_k satisfying $|f'(x)| < 1$ and containing a fixed point x_k .

2 Application on real functions

In this section we will apply the Basin Detection Method (BDM) to 3 particularly interesting real function. $\cos(x)$ is a simple function, which we will use to test our procedure. Also, we will see how different is its recursive behavior from $\cos(2x)$. The linear rational functions, are chosen as they "preserve the cross-ratio structure"[5] useful to find their convergence. And the exponential function will help us to transition to the main goal of this paper: the infinite tetration convergence.

2.1 Example 1: cosinus

let's look at the first function example: $f(x) = \cos(x)$.

Application of the BDM

1. The domain is $\mathbb{R}$ and the range is $[0, 1]$ which is continuous and satisfies $R \subsetneq D$.

Therefore, we can apply the refinement by shrinking D to be equal to R . This will increase the basin of attraction.

2. As $f'(x) = -\sin(x)$ and $|\sin(x)| = |\sin(x)|$, the range of

$f'(x) = [\sin(0), \sin(1)] \approx [0, 0.841...]$. Therefore, $|f'(x)| < 1$ for all $x \in D_2$. As $\cos(x) = x$ is not possible to solve precisely, we can look if $y = x$ and $y = \cos(x)$ intersect graphically on the figure 3.

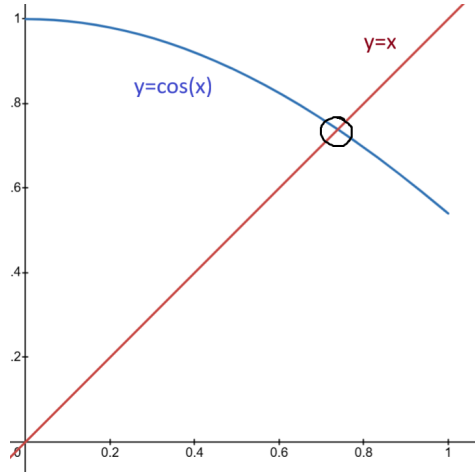


Figure 3: Intersection between $y = x$ and $y = \cos(x)$.

3. D_2 is made up of a single interval, therefore $\cos(x)$ converges to the single attracting point.

The solution to $\cos(x) = x$ does not exist in a closed form but it is approximately $x_0 = 0.739...$ and it is called the Dottie number [11].

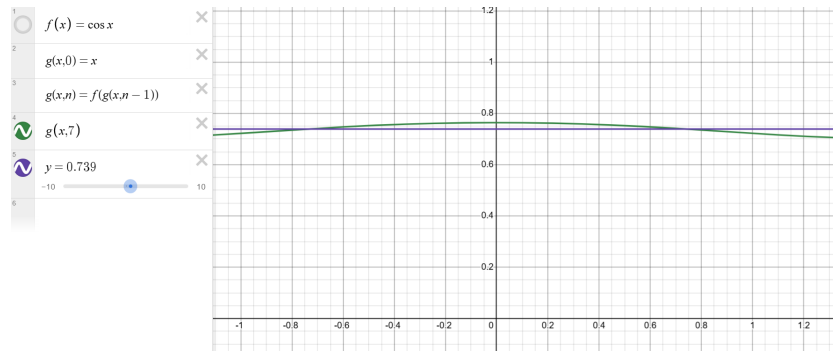


Figure 4: Dottie constant convergence

Repeating the same methodology for $g(x) = \cos(2x)$ we get to see how important is the gradient condition for an attracting point 1. We can see graphically that the only fixed point of $g(x)$ is not attracting, as its slope is bigger than 1. Therefore, there is no point of attraction for $g(x)$. let's plot $f^7(x)$ and $g^7(x)$ on the figure 5 to observe the phenomena. Note that a single horizontal contraction completely reverses the behavior of the reiterate function. This could indicate us that the boundary between stability and disorder is very tight. The smallest modification can become the distinction between perfection and chaos.

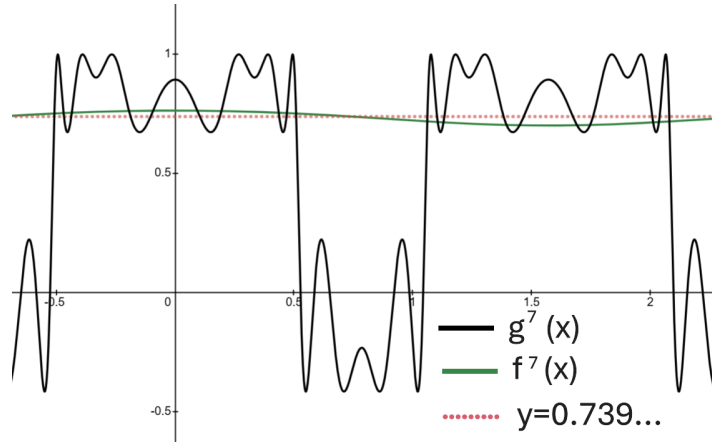


Figure 5: $\cos(x)$ vs $\cos(2x)$

2.2 Example 2: linear rational functions

let's look at another interesting family of functions. $f(x) = \frac{ax+b}{cx+d}$ where $a, b, c, d \in \mathbb{R}^+$ and $ad > bc$. The general case just requires looking at another symmetrical case. let's first understand the asymptotic behavior of this function:

$$\lim_{x \rightarrow (-\frac{d}{c})^-} f(x) = \frac{\frac{bc-ad}{c}}{0^-} = +\infty,$$

$$\lim_{x \rightarrow (-\frac{d}{c})^+} f(x) = \frac{\frac{bc-ad}{c}}{0^+} = -\infty.$$

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{a + \frac{b}{x}}{c + \frac{d}{x}} = \frac{a}{c}.$$

let's compute the derivative to understand the monotonicity:

$$f'(x) = \frac{a(cx+d) - c(ax+b)}{(cx+d)^2} = \frac{ad-bc}{(cx+d)^2}$$

As $ad > bc$, $f'(x) > 0$. So the function is monotonously increasing from a/c to $+\infty$ on the interval $(-\infty, -\frac{d}{c})$ and then monotonously increasing from $-\infty$ to a/c on the interval $(-\frac{d}{c}, +\infty)$ with a vertical asymptote at $x = -d/c$. We can graph our conclusions as shown on the figure 6.

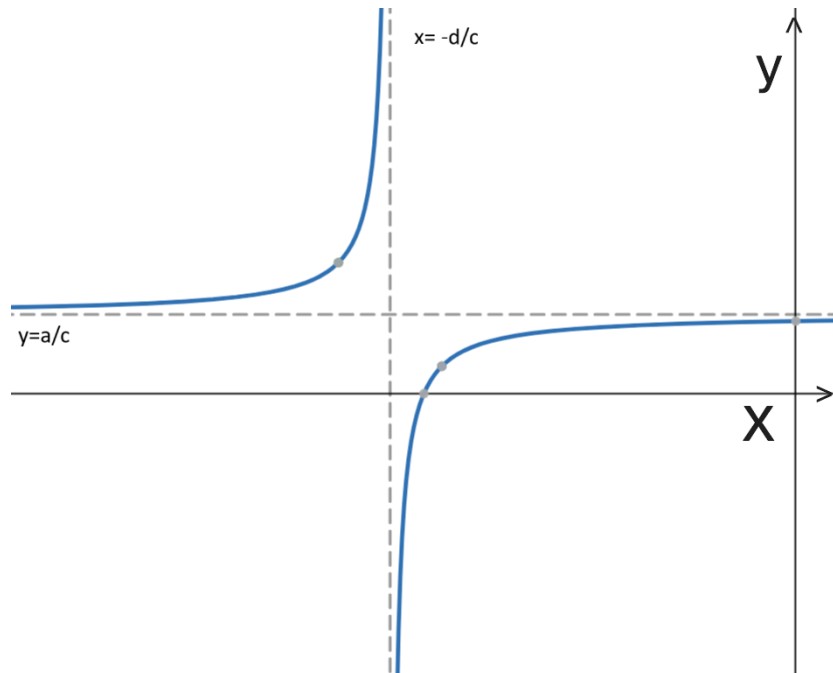


Figure 6: Note: Graph of $f(x)$

Application of the BDM

Now, let's carefully apply our BDM taking into account the asymptotic behavior of the function.

1. As f has a vertical and a horizontal asymptote, D and R are infinitely big, there is no contraction so we cannot apply the refinement.
2. let's solve $|f'(x)| = \left| \frac{ad - bc}{(cx + d)^2} \right| < 1$. As $(cx + d)^2 \geq 0$,

$$|ad - bc| < (cx + d)^2 \implies \sqrt{|ad - bc|} < |cx + d|.$$

To solve this inequality we can look at both absolute value cases separately:

1. $cx + d \geq 0 \implies \sqrt{|ad - bc|} < cx + d \implies \frac{\sqrt{|ad - bc|} - d}{c} < x$
 2. $cx + d < 0 \implies \sqrt{|ad - bc|} < -(cx + d) \implies -\sqrt{|ad - bc|} > cx + d$
 $\implies x < \frac{-\sqrt{|ad - bc|} - d}{c}$
- Therefore, the answer is the union of the 2 possibilities:

$$D \in \left(-\infty, \frac{-\sqrt{|ad - bc|} - d}{c} \right) \cup \left(\frac{\sqrt{|ad - bc|} - d}{c}, +\infty \right).$$

let's call I_1 and I_2 the first and the second intervals respectfully. As we may notice, D can be interpreted as $(-\infty, \frac{-d}{c} - k) \cup (\frac{-d}{c} + k, +\infty)$, for a constant k . This is logical as the derivative explodes to infinity around $x = -d/c$ and the intervals are symmetrical with respect to the vertical asymptote. Let $P_1 = \frac{-d}{c} - k$ and $P_2 = \frac{-d}{c} + k$. The figure 7 represents the function after restricting the domain to D .

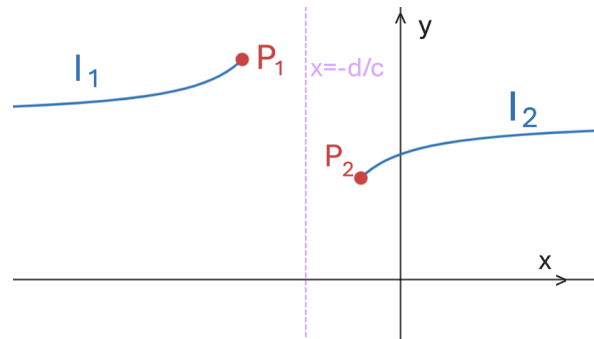


Figure 7: The function on domain D after applying the slope condition 3

3. Now let's solve $f(x) = x$:

$$\frac{ax + b}{cx + d} = x \implies ax + b = x(cx + d).$$

After expanding the right-hand side and simplifying the expression we obtain:

$$ax + b = cx^2 + dx \implies 0 = cx^2 + dx - ax - b = cx^2 + (d - a)x - b.$$

As this is a quadratic equation we can solve it to obtain 2 solutions. First let's obtain the discriminant:

$$\Delta = (d - a)^2 - 4(c)(-b) = (d - a)^2 + 4bc.$$

Applying the quadratic formula,

$$x = \frac{a - d \pm \sqrt{(d - a)^2 + 4bc}}{2c}.$$

So $f(x)$ has 2 fixed points:

$$\begin{cases} x_1 = \frac{a - d + \sqrt{(d - a)^2 + 4bc}}{2c} \\ x_2 = \frac{a - d - \sqrt{(d - a)^2 + 4bc}}{2c} \end{cases}$$

let's verify whether the fixed points are attracting by analyzing their slope:

$$f'(x_1) = \frac{ad - bc}{(cx_1 + d)^2}, \quad f'(x_2) = \frac{ad - bc}{(cx_2 + d)^2}.$$

Now, recalling the symmetric fixed points property [5], we can deduce a wonderful

relation between them:

$$\begin{aligned}
(cx_1 + d)(cx_2 + d) &= \left(\frac{a + d + \sqrt{(d-a)^2 + 4bc}}{2} \right) \left(\frac{a + d - \sqrt{(d-a)^2 + 4bc}}{2} \right) \\
&= \frac{(a + d)^2 - (\sqrt{(d-a)^2 + 4bc})^2}{4} = \frac{(a + d)^2 - ((d-a)^2 + 4bc)}{4} \\
&= \frac{(a + d)^2 - (d-a)^2 - 4bc}{4} = ad - bc.
\end{aligned}$$

$$\Rightarrow f'(x_1) \cdot f'(x_2) = \frac{ad - bc}{(cx_1 + d)^2} \cdot \frac{ad - bc}{(cx_2 + d)^2} = \frac{(ad - bc)^2}{(ad - bc)^2} = 1 \Rightarrow \frac{1}{f'(x_1)} = f'(x_2).$$

Therefore, we have 2 fixed points and their derivatives are reciprocal. Notice that P_1 and P_2 were the 2 points with the slope equal to 1. Although the proof is beyond this paper, f is symmetrical with respect to the line P_1P_2 . As P_1P_2 is perpendicular to the line $y = x$, x_1 and x_2 are symmetrical with respect to P_1P_2 . This is the intuitive way to see why the fixed points are reciprocal. The idea is shown on the figure 8.

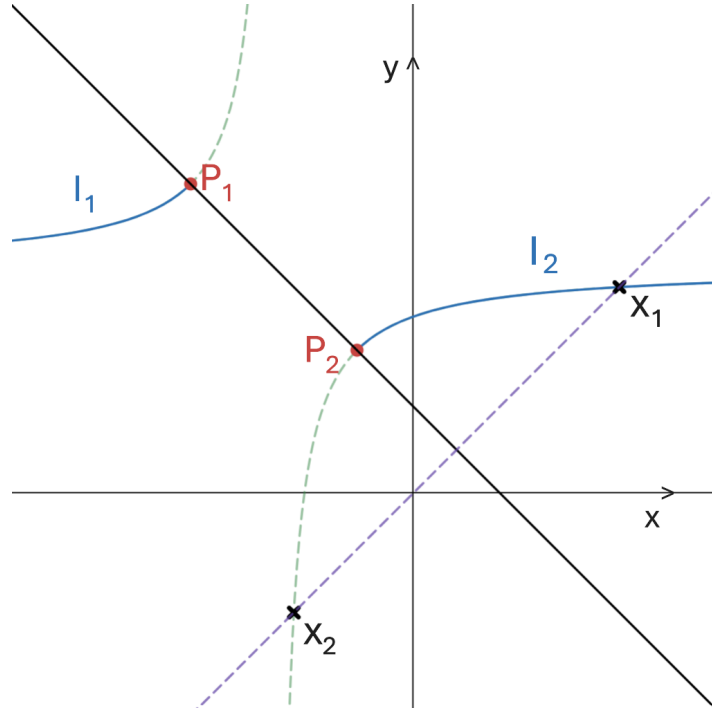


Figure 8: Symmetry with respect to the line P_1P_2 .

As both derivatives are greater than 0, one must be greater than 1 and the other one

must be smaller than 1. We can conclude that one of them is an attracting point and the other one is not. As $x_1 > x_2$,

$$0 < f'(x_1) = \frac{ad - bc}{(cx_1 + d)^2} < f'(x_2) = \frac{ad - bc}{(cx_2 + d)^2}.$$

Hence, x_1 is the attracting point.

$$\begin{aligned} x_1 &= \frac{a - d + \sqrt{(d - a)^2 + 4bc}}{2c} = \frac{a - d + \sqrt{d^2 - 2ad + a^2 + 4bc}}{2c} \\ &> \frac{a - d - \sqrt{d^2 - 2ad + a^2 + 4ad}}{2c} = \frac{a - d + \sqrt{(d + a)^2}}{2c} \\ &= \frac{a - d + (d + a)}{2c} = \frac{a}{c} > -\frac{d}{c} \end{aligned}$$

As x_1 is located on the right of the vertical asymptote, $x_1 \in I_2$.

4. For this step we recall $D = I_2$ as I_1 does not have any fixed point. Any $x \in I_2$ will converge to x_2 in the limit.

2.3 Example 3: exponential function

Now we will proceed to analyze the exponential family of functions. The study of these functions will help us finding the convergence of power towers. To clarify, we want to solve $a^{a^{a^{\dots}}} = {}^\infty a$, $a \in \mathbb{R}^+$. Let $x_0 = {}^\infty a$ in the case that ${}^\infty a$ converges to a number. It is obvious, that for big values of a , x_0 diverges to $+\infty$. For example, ${}^\infty 2 \rightarrow \infty$. But for $a = 1$, $x_0 = {}^\infty 1 = 1$. If for $a = 1$ the tower converges and for $a = 2$ the tower diverges, there is clearly a number $p \in (1, 2)$ such that for $a \in (1, p]$, ${}^\infty a$ converges and for $a \in (p, 2]$, ${}^\infty a$ diverges. As ${}^\infty 0$ is undefined, it will be also interesting to see what happens to the convergence as a approaches 0. let's define $f(x) = a^x$. Notice that $f^n(x)$ as $n \rightarrow \infty$ is exactly the power tower we want to analyze. If a power tower converges to x_0 , then x_0 must be an attracting point. Now let's apply the BDM to find the attracting points and their basins

of attraction. The basin of attraction will mean the following: for what values b around x_0 is it true that $b^{a^{a^{\dots}}} \rightarrow x_0$.

Convergence of power towers

In this case, we will apply BDM without specifying the steps, as they are too long. We will consider the basin of attraction at the end, when the attracting point is already found. This will optimize the calculus. To find the attracting points, we need to solve the following system of equations:

$$\begin{cases} f(x_0) = x_0 \\ |f'(x_0)| < 1 \end{cases}$$

For all the values of a satisfying this system, the infinite tetration ${}^\infty a$ converges to the attracting point x_0 . It is known that $f'(x) = \ln(a)a^x$. Substituting,

$$\begin{cases} a^{x_0} = x_0 \\ |\ln(a)a^{x_0}| < 1 \end{cases} \implies a = x_0^{\frac{1}{x_0}}.$$

Substituting into the second equation:

$$\left| \frac{\ln(x_0)}{x_0} (x_0^{\frac{1}{x_0}})^{x_0} \right| = \left| \frac{\ln(x_0)}{x_0} x_0 \right| = |\ln(x_0)| < 1.$$

By definition of the absolute value,

$$|\ln(x_0)| < 1 \iff -1 < \ln(x_0) < 1.$$

As f is an exponential, it is strictly increasing on the positive domain. Therefore, we can calculate the values that the attracting point can take: $x_0 \in (e^{-1}, e)$. Although we know $a(x_0)$, it would be much more representative to have the inverse, the general $x_0(a)$.

Lambert function

We need introduce a more advanced equation solving technique in order to find this general solution. From the first equation we had $a^{x_0} = x_0$. x_0 is the multiplicative factor and the exponent in the same equality. This type of equations can sometimes be solved using the lambert function [3]. By definition, $W(xe^x) = x$. Therefore, the lambert function is the inverse function of the function $u(x) = xe^x$. For example, if we had $xe^x = 3$, $x = W(3)$ is a valid solution.

General form for positive power towers [4]

If we remember the equality from our first equation, $a^{x_0} = x_0$ We can take the logarithm as both sides are greater than 0. $\frac{\ln(x_0)}{x_0} = \ln(a) \implies \frac{-\ln(x_0)}{x_0} = -\ln(a)$. Now, we can make a clever substitution: $u = \ln(x_0)$, so $\frac{-u}{e^u} = -\ln(a) = -ue^{-u}$. We can now apply the Lambert function [3] on both sides: $W(-\ln(a)) = W(-ue^{-u}) = -u$. Substituting back, $\ln(x_0) = -W(-\ln(a))$. Finally,

$$\boxed{x_0(a) = e^{-W(-\ln(a))}} \quad (4)$$

We have deduced a general formula for the infinite positive power towers. The graph 9 shows the value x_0 as the result of the infinite tetration: $x_0(a) = {}^\infty a$.

As the function is monotonously increasing, the inverse is also monotonously increasing [9]. For this reason, The lower bound is

$$a(\min(x_0)) = a(e^{-1}) = e^{\frac{\ln(e^{-1})}{e^{-1}}} = e^{\frac{-1}{e^{-1}}} = e^{-e}.$$

And the upper bound is

$$a(\max(x_0)) = a(e) = e^{\frac{1}{e}}.$$

The power towers ${}^\infty a$ converges on a very specific interval with the bounds tightly related to e . Recursion has simplified an infinitely long problem to a problem asking to find an

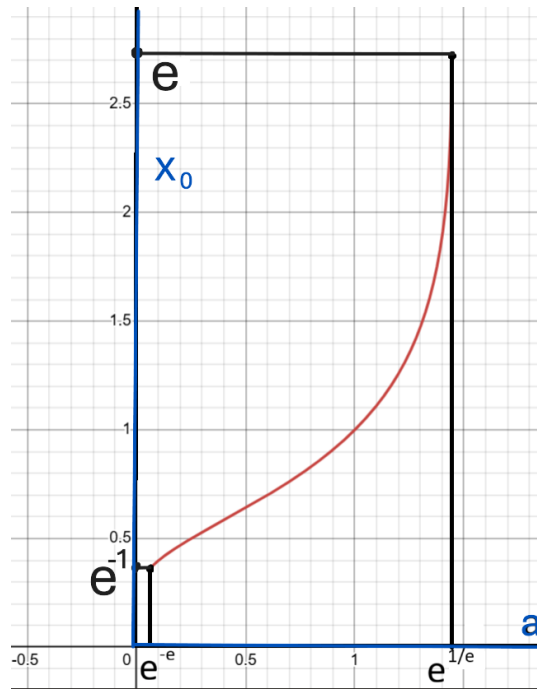


Figure 9: Convergence of infinite tetration

attracting point. Not only this, but we can now find the basin of attraction around x_0 . When does $f^n(x)$ converge. The meaning of power towers for this is the following: When does $a^{a^{\dots a^b}} \rightarrow x_0$? Let's find this interval applying the condition 3.

$$f'(x) = \ln(a)a^x \implies |\ln(a)a^x| < 1.$$

Observe that $a^x > 0$ as $a > 0$. Therefore the sign of the value inside the absolute value depends on the sign of $\ln a$. When $a > 1$, $\ln a \geq 0$ and when $a < 1$, $\ln a < 0$. Now we can get rid of the absolute value analyzing 2 different cases:

$$\begin{cases} a > 1 \implies \ln(a)a^x < 1 \implies a^x < \frac{1}{\ln(a)}, \\ a < 1 \implies -\ln(a)a^x < 1 \implies a^x < \frac{1}{-\ln(a)} \end{cases}$$

$$\implies \begin{cases} \ln(\ln(a)) < -x \ln(a) \implies x < \frac{-\ln(\ln(a))}{\ln(a)}, \\ \ln(-\ln(a)) < -x \ln(a) \implies x > \frac{-\ln(-\ln(a))}{\ln(a)}. \end{cases}$$

In summary, when $a \in (1, e^{1/e})$, $a^{a^{\cdots a^b}}$ converges to $x_0(a) = e^{-W(-\ln(a))}$ for

$$b \in \left(-\infty, \frac{-\ln(\ln(a))}{\ln(a)} \right).$$

And when $a \in (e^{-e}, 1)$, $a^{a^{\cdots a^b}}$ converges to $x_0(a) = e^{-W(-\ln(a))}$ for

$$b \in \left(\frac{-\ln(-\ln(a))}{\ln(a)}, +\infty \right).$$

3 Application on complex functions

In this section, we will analyse the convergence of the complex power towers. The conditions 2 and 3 also hold for complex numbers, but the proof is beyond our goals. It is coherent to note that $|z| = \sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2}$ when applying $|f'(z)| < 1$.

Multiplicity of answers

Once we leave the real world we face the problem of infinitely many solutions. To begin, let's solve the simplest complex tetration in order to illustrate the point: $z = {}^2i = i^i$.

$$i = e^{i(\pi/2+2\pi k)} \implies z = e^{i(\pi/2+2\pi k) \cdot i} = e^{-(\pi/2+2\pi k)} = \boxed{e^{-\pi/2} e^{-2\pi k}}$$

For different values of k we get infinitely many solutions. This was only a power tower of height 2. And we want to find a power tower of infinite height, so there will be even more solutions. For convenience we will only handle the single solution for $k = 0$.

3.1 Attracting points on complex tetration

let's define the new problem. We have a complex function $f(z) = b^z$ where $b, z \in \mathbb{C}$. And we want to find out when does $f^n(z)$ converge as $n \rightarrow +\infty$. Therefore, we want to discover the family of functions $f(z)$ that have at least 1 attracting point z_0 . For that purpose we should

apply the 2 conditions that attractive points satisfy:

$$\begin{cases} f(z_0) = z_0 \\ |f'(z_0)| < 1 \end{cases} \implies \begin{cases} b^{z_0} = z_0 \\ |\ln(b)b^{z_0}| < 1 \end{cases}$$

Plugging in the first equation into the second we get $|\ln(b)z_0| < 1$.

Branches of the Lambert function [3]

In order to find $z_0(b)$, we need to clarify another important aspect. We will need to apply the Lambert function to a complex number. The lambert function satisfies $W(ze^z) = z$. As we have previously seen that the exponential complex numbers have infinitely many solutions. For this reason, the lambert function has infinitely many branches W_k on the complex domain. As we have fixed $k = 0$ at the section 3, we will only consider the branch 0 of the lambert function: $W_0(x)$.

Complex results of tetration

Surprisingly, the complex dimension does not add any difficulty into finding the expression $z_0(b)$, and we can derive it the same way as equation 4, with b instead of a and z_0 instead of x_0 : $z_0(b) = e^{-W(-\ln(b))}$

We can graph the results in 3D. The x and y dimensions will represent the real and the imaginary parts respectively of the complex constant b . And the dimension z will show $|z_0|$. We know that

$$z_0(b) = e^{-W(-\ln(b))} \implies |z_0| = |e^{-W(-\ln(x+yi))}|$$

But we cannot forget the slope condition 3. Hence, we must restrict the domain of this function to $|\ln(z_0)z_0| < 1$. let's introduce the function The plot is displayed on the figure 10. The orange curve represents $x_0(a)$ that we found for the real values.

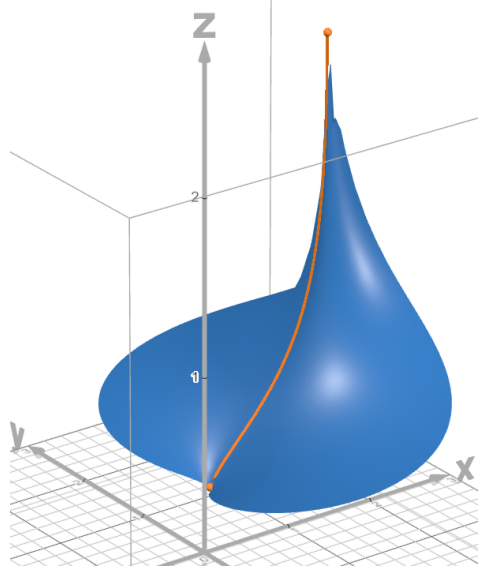


Figure 10: 3D view of the convergence graph

As we have done for the real power towers, let's try finding the basin of attraction for a function $f(z) = b^z$ where $z = c + di$. The general case is beyond the scope of this investigation, so let's analyze the particular case $b = i$. To begin, let's simplify the derivative:

$$\begin{aligned} f'(z) &= \ln(b)b^z = \ln(i)i^z = \ln(e^{i\pi/2})i^{c+di} = \frac{i\pi}{2}i^c i^{di} \\ &= \frac{i^{c+1}\pi}{2}e^{\frac{i\pi}{2}di} = \frac{e^{(c+1)\ln(i)}\pi}{2}e^{\frac{-\pi}{2}d} = \frac{e^{i\frac{\pi}{2}(c+1)}\pi}{2}e^{\frac{-\pi}{2}d}. \end{aligned}$$

To find the basin of attraction, we must solve $|f'(z)| < 1$. Hence,

$$\left| \frac{e^{i\frac{\pi}{2}(c+1)}\pi}{2}e^{\frac{-\pi}{2}d} \right| = \frac{\pi}{2}e^{\frac{-\pi}{2}d}|e^{i\frac{\pi}{2}(c+1)}| = \frac{\pi}{2}e^{\frac{-\pi}{2}d}, \text{ as the modulus is 1.}$$

Therefore, the module of the derivative does not depend on c . This means that any complex c is valid. Now, let's look when is it smaller than 1:

$$\frac{\pi}{2}e^{\frac{-\pi}{2}d} < 1 \implies e^{\frac{-\pi}{2}d} < \frac{2}{\pi} \implies \frac{-\pi}{2}d < \ln\left(\frac{2}{\pi}\right) \implies d > -\frac{2}{\pi}\ln\left(\frac{2}{\pi}\right) \approx 0.287.$$

To summarize, the complex infinite power tower $b^{b^{\dots b^{c+di}}}$ converges to $z_0(b) = e^{-W(-\ln(b))}$ when $d > -\frac{2}{\pi}\ln\left(\frac{2}{\pi}\right)$.

4 Conclusion

In this paper we have achieved our goal, exploring the recursive behavior of the functions in the limit. Answering the research question, we discovered that a single attracting point is necessary and sufficient for a function to converge. We came up with a general method to apply to any function and find its attracting points and basins of attractions. This process was hard to deduce in a general way to include all type of functions, as some functions could have multiple attracting points simultaneously. This implies a restriction to study the local instead of global convergence. Then we applied these conditions on real functions where we used a handy domain contraction idea to maximize the basin of attraction.

The exponential function led us to discover which of the power towers converged applying the recursive thinking. Finally, we took an insight into the complex tetration, which would be impossible to deal with without the conditions we deduced for real functions. We found out that our conditions continued working for complex numbers, although we had to watch out for the multiplicity of answers. On our way we had to introduce the lambert function and famous theorems as the Mean Value Theorem and the Banach Fixed Point Theorem. It was also very helpful to plot the resulting graphs in Desmos as a visual support for recursive functions. This way we could clearly see an intuitive demonstration for the algebraic equalities.

Although we have found a convergence range for rational functions, we did not prove convergence on all domain. This is a goal for a future paper. We could also find a precise formula describing the basin of attraction for any complex exponential function.

We have explored the functions at their limits, and found out that many functions do have intervals or regions that behave in a stable way at their recursive limit. A single attracting point is necessary for the complex power tower to converge.

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Appendix A: Proof of Claim 1

This is based on a proof by Conrad K [2]. Let's zoom in and watch the local behavior of $f(x)$ near a fixed point x_0 . By definition, $f(x_0) = x_0$. Now let's observe values very close to x_0 :

$$z = x_0 + \epsilon, \quad \epsilon \neq 0, \quad \epsilon \text{ small.}$$

In order for x_0 to be an attracting point, it must have a basin of attraction. For that reason, there exists $\delta > 0$ such that, for all $0 < \epsilon < \delta$,

$$|f(x_0 + \epsilon) - f(x_0)| < |\epsilon|. \quad (5)$$

As $f(x)$ is differentiable at x_0 , let's find the Taylor expansion at this point:

$$f(x_0 + \epsilon) = f(x_0) + f'(x_0)\epsilon + \frac{1}{2}f''(x_0)\epsilon^2 + \dots = f(x_0) + f'(x_0)\epsilon + O(\epsilon^2).$$

Operating,

$$f(x_0 + \epsilon) - f(x_0) = \epsilon(f'(x_0) + O(\epsilon)).$$

To analyze the distance between $f(x)$ and $f(x_0 + \epsilon)$ let's take the absolute value:

$$|f(x_0 + \epsilon) - f(x_0)| = |\epsilon| \cdot |f'(x_0) + O(\epsilon)|.$$

By the condition (5) for an attracting point,

$$|\epsilon| |f'(x_0) + O(\epsilon)| < |\epsilon| \implies |f'(x_0) + O(\epsilon)| < 1.$$

As $\epsilon \rightarrow 0$, $O(\epsilon) \rightarrow 0$. So the final condition is,

$$\boxed{|f'(x_0)| < 1} \quad (6)$$

Q.E.D.